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How do personality traits of college students affect their learning flow experience?

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ABSTRACT

Researchers hypothesize that awareness of one's learning motivations leads to diverse levels of learning flow experience depending on one's personality. The present study examined first the personality trait associated with the highest learning flow experience and whether students' learning motivation mediates the relationship between their different personality traits and level of learning flow experience. The sample consists of 450 undergraduate students (257 women and 193 men) from China, and the data were collected using online questionnaires forms. Through correction analysis and stepwise regression analysis, researchers determined in the first step that the personality with the highest degree of learning flow experience was Agreeableness. In a second step, a test of mediating effects was conducted to examine how this personality trait influences the learning flow experience, directly or indirectly, via various coping styles (intrinsic or extrinsic motivation). The results indicated that agreeableness directly predicted the degree of immersion learning experience and influenced the learning flow experience through intrinsic learning motivation. It highlights the significance of college students' motivation to learn as a mediator of the link between the agreeableness personality trait and the learning flow experience.

1. Introduction

Learning is a multifaceted and intricate process influenced by a range of factors, including the big five personality traits (Cao & Meng, 2020; Entwistle & Ramsden, 2015; Marcela, 2015), learning motivation (Aguilera-Hermida, 2020; Cao & Meng, 2020; El-Adl & Alkharusi, 2020) and learning flow experience (Aguilera-Hermida, 2020; Kiili et al., 2012). Within the framework of learning theory, learning encompasses various dimensions and involves behavioral shifts that reflect both internal and external motivational changes. It is an ongoing process that facilitates the development and refinement of individuals' personalities. Learning is a dynamic phenomenon as individuals construct knowledge and attribute meaning to their experiences. Learning theory provides a fundamental framework for comprehending the intricate mechanisms of knowledge acquisition and retention. According to this theoretical perspective, personality traits, sense of immersion experiences, and behavioral processes interact with the environment to shape individuals' learning experiences and ultimately determine the effectiveness of knowledge acquisition (Lombardi et al., 2021). Personality traits play a significant role in learning theory as they influence learning motivation, preferences, behaviors, and outcomes (McCrae & Costa, 1995; Obrenovic et al., 2022).Furthermore, Learning motivation plays a crucial role in driving individuals to engage in learning activities and

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pursue educational goals (Bailey et al., 2021; Hidi & Harackiewicz, 2000).

Recent research has also emphasized the significance of learning flow experiences (Wang et al., 2020), highlighted the significance of learning flow experiences, with some scholars exploring the correlation between personality traits and learning flow experience (Marty-Dugas & Smilek, 2019); others have explored the relationship between learning motivation and learning flow experiences (Dunkel et al., 2022; Kazakova & Shastina, 2019; Meng et al., 2016; Wang et al., 2020) ; Despite the acknowledged importance of personality traits, learning motivation, and learning flow experiences in shaping student learning outcomes, the intricate dynamics and combined effects of these factors remain under-explored, particularly in countries with diverse education systems (Hennebry-Leung & Xiao, 2023; Monroy & Hernandez Pina, 2014) , Different student background characteristics and learning motivation predict different learning outcomes (Lin, 2020). Therefore, this study attempted to address this research gap by investigating Chinese college students' overall personality traits, learning motivation, and learning flow experiences and examining how these factors interact. First, we identified the personality traits that most influence learning flow experiences among Chinese college students, and second, based on learning and motivation theories, we used a well-established motivation measurement scale with Chinese characteristics to explore the most obvious learning motivation items that best serve as mediating variables between personality traits and learning immersion experiences.

Therefore, this study aims to explore the dynamic relationship between university students' personality traits, learning motivation, and learning flow experiences to gain insights into the dynamics that shape their educational journey. By understanding these dynamics, Chinese educators and researchers can develop targeted interventions and teaching strategies to increase student engagement and motivation, fostering more effective and enjoyable learning experiences. Additionally, by contributing to the existing knowledge of learning theory with a specific focus on the Chinese context, the findings of this study have the potential to shape educational practices and provide valuable insights for creating optimal learning environments that promote motivation, learning flow experiences, and learning process among university students with the Big Five personality traits.

2. Theory foundation and hypotheses

2.1. Personality traits and learning motivation

Motivation is the source of student learning and a catch-all term for the diverse needs that drive students to learn (Filgona et al., 2020). These needs reflect the external demands of family and society on students' learning and the internal demands of students to pursue their passions and realize their life goals. Learners' motivation propels them to achieve their learning objectives; therefore, it is likely the most crucial factor in learning (Borah, 2021);learners must be interested in learning. Otherwise, the education system's efforts will fail. Therefore, understanding and developing knowledge of the factors influencing students' motivation to learn can improve their academic performance(Gbollie & Keamu, 2017). There is evidence that personality variables also play a role in learning motivation (O'Connor & Paunonen, 2007), although ability and intelligence are considered influential. Additionally, the literature suggests that personality traits affect learning motivation (Costa & McCrae, 1992; Greven et al., 2008; Kircaburun et al., 2020). Komarraju and her colleagues examined the relationship between the Big Five personality traits and learning motivation to gain insight into students' learning styles (Komarraju & Karau, 2005; Komarraju et al., 2009). For instance, Komarraju and Karau (2005) analyzed 172 students using the Big Five Personality Inventory (NEO-FFI) and the Learning Motivation Questionnaire (AMI) as research instruments. The researchers discovered a correlation between personality traits and learning motivation among university students. According to the findings, the Big Five personality traits were significantly associated with all 16 AMI subscales. Commitment was explained by extraversion and openness; achievement by agreeableness, neuroticism, and extraversion; avoidance by neuroticism and extraversion; and avoidance was negatively related to agreeableness and openness.

In addition, Clark and Schroth (2010) used multiple regression analysis to compare three intrinsic and extrinsic motivations to five personality variables among 451 first-year college students. Students who were intrinsically motivated to attend school tended to be extroverted, pleasant, responsible, and open to new experiences; those who were extrinsically motivated tended to be extroverted, pleasant, responsible, and neurotic; and those who lacked motivation tended to be unfriendly and apathetic. Different personality types were motivated to attend school for varying motivations. Hazrati-Viari et al. (2012) selected motivated undergraduates and had them fill out the NEO-FFI Personality Questionnaire and AMS-C Learning Motivation Questionnaire. The results demonstrated that agreeableness predicted general motivation, whereas openness predicted only intrinsic motivation; the study also demonstrated that learning motivation affected learning performance. The effect on educational outcomes was moderated by Openness and Agreeableness. De Feyter et al. (2012) concluded that three trait dimensions—agreeableness, extraversion, and neuroticism—can indirectly influence learning achievement through learning motivation. Apostolov and Geldenhuys (2022) asked 285 Australian undergraduates to complete the International Personality Inventory Pool and the Motivational Strategies Learning Questionnaire to measure personality and motivation, respectively. The structural equation model results showed diligence had the most significant relationship with learning motivation, while neuroticism was negatively related. In addition, the anxiety aspect of neuroticism was the only significant positive predictor of learning motivation.

Because we aim to find a correlation between the Big Five personality traits and learning motivation. Based on the preceding discussion, the following hypothesis is developed:

Hypothesis 1. ***(H1)***: The Big Five personality correlate positively with learning motivation.

2.2. Personality traits and learning flow experiences

Flow is said to be associated with an increase in positive emotions and a decrease in negative emotions and can potentially improve well-being. Flow, or optimal experience, can be beneficial for mental health and productivity. This may be particularly important for adolescents, who are constantly faced with demands for adjustment due to the stage of development they are in (Annalakshmi et al., 2020).

According to Csíkszentmihályi, various factors impact the variances through individual flow experiences. Such as the characteristics of the activity itself and personality traits. These personality traits can explain the intensity and frequency of their flow experiences. Self-directed personality is the fundamental dispositional trait for flow experiences. Ye (2013) suggested that individuals who are frequently immersed have self-directed personalities, are self-motivated themselves, are not dependent on rewards given from outside and are more likely to experience true happiness and pleasure (Lopez-Garrido, 2021).

Kimiecik and Stein (1992) proposed a series of personality variables that influence individuals' flow experiences, such as self-confidence, perceptual-motor ability, and attentional style; it can be very useful in constructing immersive experiences. The ESM method was used by (Asakawa, 2004) to divide a sample of Japanese university students into self-directed and non-self-directed personality groups and investigated the effect of perceptions of challenge/skill on the flow experiences of both groups and how these two groups balanced their perceived challenges and skills in their daily lives. The findings confirmed that highly challenging versus highly skillful situations were more likely to produce flow experiences; that students with an autotelic personality balanced challenge/skill at a higher level than students with a non-autotelic personality; that students with an autotelic solid personality tended to perceive challenge higher than perceived skill, whereas students with a non-autotelic personality tended to keep perceived challenge/skill at a realistic level; and that Asakawa (2010) showed that Japanese university students with an introverted personality or those who were frequently immersed in their daily activities were more able to maintain a higher sense of self and lower concerns, and more often used positive coping strategies. In this study, the authors even used the frequency of flow experiences to indicate self-directed personality.

Liang (2013) studied learning flow experiences and their influencing factors among higher education students. According to the study, individuals with distinct self-directed personality tendencies had significantly diverse learning flow experiences and substantial disparities in the dimensions of learning flow experiences.; self-directed personality played a part in mediating the effects of learning self-efficacy, learning motivation, and learning environment on learning flow experiences.

Annalakshmi et al. (2020) analyzed data from a sample of 200 adolescents from Kerala and Tamil Nadu through multiple regression analysis to identify personality traits that predicted mobility experiences; the results of the multiple regression showed that extraversion was a positive predictor of overall mobility experiences and ANOVA compared the effects of gender on personality and mobility, finding that females were higher in honesty and humility, emotionalism, extraversion, self-awareness and openness to experience, as well as higher in overall mobility experience. Therefore, the following hypothesis is presented:

Hypothesis 2. ***(H2)***: Big five personality correlate positively with learning flow experience.

2.3. Learning motivation and learning flow experience

It is important for students to experience mindfulness in learning so that they can find enjoyment in learning (Cheng, 2013; Van Schaik et al., 2012), as the main function of activities that facilitate mindfulness is to provide enjoyable experiences (Csikszentmihalyi & Csikzentmihaly, 1990). Flow emphasizes the positive aspects of learning. When people are immersed in a task or activity, they can experience enjoyable moments (Tobert & Moneta, 2013), and by experiencing learning for pleasure, students may reduce anxiety, stress, boredom, or other negative states of mind (Csikszentmihalyi & Csikzentmihaly, 1990). In addition, mindful flow enhances students' self-esteem, learning motivation and learning satisfaction (Joo et al., 2011), and therefore students will have stronger motivation to learn. Conversely, a number of studies have also reported that personal motivational factors such as self-efficacy (Meece et al., 1988; Puzziferro, 2008)and intrinsic value (Jackson et al., 2008) are influential factors in learning flow.

Wang et al. (2010) used flow theory to study the state of mathematics examinations of high school students and showed that highly motivated students enjoyed the challenge and experienced more flow experiences than low motivated students. Exam flow experience and internal motivation for learning had a substantial favorable link.

Liang (2013) used the Learning Motivation Questionnaire and the Learning Flow Experience Scale for university students to examine the connections between motivation and flow experience among high education students, showing that there were significant differences in the learning flow experience gained by individuals with different learning motivations, as well as significant differences in the various dimensions of the learning flow experience. Ye (2013) explored the influence of intrinsic motivation on the flow experience of adolescents. An "A" exercise, a post-activity questionnaire, and an adolescent learning flow experience questionnaire were used as materials for 98 junior high school students. The results showed that Intrinsic motivation in the experimental group had significant positive correlations with task-skill balance and sense of control and significant negative correlations with awareness-action integration, purpose and feedback, concentration, forgetfulness, enjoyment of the experience, and overall; value evaluation had significant negative correlations with all factors of learning flow. There were no significant correlations between intrinsic motivation, value evaluation, and learning flow experience factors in the control group. According to regression analysis, intrinsic motivation has a considerable impact on the learning flow experience.

Hofferber et al. (2016) used a study of 158 students of grade six who worked on the adaptations of Harvest Mice (Micromys minutes) with living animals, it was found that compared to control teaching behaviors, autonomy-supportive teaching behaviors

resulted in students' intrinsic motivation and flow experiences. A significant difference was found in students' intrinsic motivation and mind-flow experiences as a result of autonomy-supportive teaching behaviors compared to controlled teaching behaviors.

Hong et al. (2017) found that intrinsic motivation was positively related to mind-flow experience through validated factor analysis and structural equation modelling of validated data; intrinsic motivation for Chinese language learning predicted self-efficacy for online learning. Therefore, the following hypothesis is provided:

Hypothesis 3. ***(H3)***: College students' learning motivation correlate positively with the learning flow experience.

Hypothesis 4. ***(H4)***: College students' learning motivation mediates different personality traits in the learning flow experiences.

2.4. The present study

Flow theory has been extensively investigated to explore individuals' attitudes, behaviors, and experiences in various contexts (da Silva deMatos, de Sa, & de Oliveira Duarte, 2021; Kang, Lee, & Namkung, 2018). The concept of learning flow experience not only has applications in the field of learning but also manifests as a common phenomenon in everyday learning. It has the potential to enhance learning efficiency, promote psychological well-being, and influence personality development. Flow experiences in mediated environments can be observed when individuals engage in activities such as reading a book, watching a film, or playing a computer game. Previous research consistently demonstrates that flow experiences are characterized by a strong sense of positivity and satisfaction (Weibel et al., 2010). However, recent scholarship has focused on exploring how these traits influence experiences of learning flow (Dunkel et al., 2022; Lin, 2020), leaving research gaps regarding the potential mediating role of learning motivation, particularly in different educational systems and across various personality types. Therefore, the present study aims to examine the relationship between Big Five personality traits, learning motivation, and flow learning experiences among college students in China. Additionally, drawing on learning theory, we aim to highlight the potential mediating role of learning motivation in this specific context.

3. Material and methods

3.1. Participants

The study collected data from a random sample of university students across China, with 470 participants who completed the three different survey questionnaires. After eliminating invalid questionnaires, 450 participants were included finally, resulting in a 95.74 % effective rate.

Table 1 presents an overview of the characteristics of the participants. The sample consisted of 57.11 % (N = 257) female and 42.89 % (N = 193) male students. Most participants, 59.78 % (N = 269), were juniors or seniors, while 40.22 % (N = 181) were fresh man or sophomore. Among the participants, 40.22 % (N = 181) were majoring in arts and history, while 56.44 % (N = 254) were majoring in science and engineering. Moreover, 56.44 % (N = 254) of the participants were from urban families, whereas 43.56 % (N = 196) were from rural families. Regarding family structure, 53.33 % (N = 240) were only children, while 46.67 % (N = 210) were non-only children. Finally, the monthly income of participants' households varied, with 18 % (N = 81) of the participants earning 10,000 RMB or more per month, while the majority, 82 % (N = 360), were from households with a monthly income of less than 10,000 RMB. The sample was thus diverse in terms of gender, grade level, major, family background, and income, making it representative of the broader university student population in China.

Table 1
Demographic statistics (N = 450).

Variables	Frequency	Percent (%)
Gender		
Male	193	57.1
Female	257	42.9
Grade		
Freshman/Sophomore	181	40.2
Junior/Senior	269	59.8
Major		
Arts and History	196	43.6
Science and Engineering	254	56.4
Family Location		
Urban	254	56.4
Rural	196	43.6
Only child or not		
Yes	240	53.3
No	210	46.7
Family income		
10,000RMB or more	81	18.0
10,000RMB or less	369	82.0

3.2. Questionnaire design and Measures

Three questionnaires were utilized in this study to measure learning motivation, personality traits, and learning flow experience. For a full description of the corresponding factors and the content of the constituent questions, please see "Appendix-1: Questionnaire"; Table 1 shows the reliability values and the number of question items for each factor..

3.2.1. Big Five Personality Questionnaire

We administered the Chinese Big Five Personality Inventory - Short Version (CBF-PI-B) developed by Kong & Wang, 2021 to evaluate personality traits. We administered the Chinese Big Five Personality Inventory - Short Version (CBF-PI-B) developed by Kong & Wang, 2021 to evaluate personality traits. The questionnaire consists of 40 items categorized into five traits: Neuroticism, Conscientiousness, Agreeableness, Openness, and Extraversion. Participants rated each item on a six-point Likert scale. "The reliability of all 40 items showed robust internal consistency, with a Cronbach's alpha coefficient of 0.85 and a KOM value of 0.92, indicating strong reliability.

3.2.2. Learning Motivation Questionnaire

The revised Learning Motivation Scale (RLMS), validated for college students by Chi and Xin (2006), was employed to measure learning motivation. The scale comprises 26 items categorized into internal (14 items) and external (12 items) motivation. Participants rated each item on a six-point Likert scale. The overall scale demonstrated good reliability, with a Cronbach's alpha coefficient of 0.91 and a KOM value of 0.90.

3.2.3. Learning Flow Experience Questionnaire

We used the Youth Learning Flow Questionnaire, developed by Ye (2013), to assess the flow experience of learning. The questionnaire consists of 50 items categorized into seven dimensions. Participants rated each item on a six-point Likert scale. The overall scale exhibited high internal consistency, with a Cronbach's alpha coefficient of 0.96 and a KOM value of 0.90.

3.3. Research procedure

This study was conducted in three stages to ensure the accuracy and reliability of the findings. The first stage involved the translation of the questionnaire. The chosen translation strategy was back translation, which involved multiple steps. Initially, the researcher translated the questionnaire from its original language to Chinese. Subsequently, a bilingual individual with a doctorate degree, proficient in both Chinese and English, translated the Chinese version back into the original language, without altering the original scale. The suitability of the Chinese questions was then discussed with a bilingual expert from mainland China. This iterative process was repeated once more to finalize the English and Chinese versions of the questionnaire.

The second phase of the research involved the selection, administration, and analysis of the Chinese student sample. Convenience sampling was primarily used to select the sample between August 1 and 30 August 30, 2021, which was administered to groups at different school levels. Additionally, data were collected from students at various levels of study to facilitate subsequent analysis and comparison.

Next, *The third stage comprised data analysis and report result preparation. Initially*. As shown in Fig. 1 , To achieve the research objectives, we divided the study into two phases of exploration. The first step, we identified the five personality traits that exert the most influence on the learning flow experience of college students. Building upon learning theory, we incorporated learning motivation variables to examine if learning motivation could act as a mediating variable, specifically assessing the most mediated

Table 2
Cronbach's Alpha and KOM Value of Three Questionnaires and Sixteen Variables.

Three questionnaires and sixteen variables			Items numbers		Cronbach's alpha		KOM value	
Big five personality		*X1. Neuroticism.*	8	40	**0.78**	0.85	**0.77**	0.92
		X2. Consciousness	8		**0.76**		**0.84**	
		X3. Agreeableness	8		**0.81**		**0.89**	
		X4. Openness	8		**0.83**		**0.86**	
		X5. Extraversion	8		**0.72**		**0.77**	
Learning motivation	**B1 : Intrinsic Learning Motivation**	m1 : Challenging:	8	14	**0.81**	0.91	**0.83**	0.90
		m2 : Enthusiasm	6		**0.73**		**0.8**	
	B2 : Extrinsic Learning Motivation	m3 : Choosing simple tasks	5	12	**0.70**		**0.7**	
		m4 : Relying on others to evaluate.	2		**0.73**		**0.5**	
		m5 : Seeking rewards	5		**0.72**		**0.74**	
Learning flow experience	**C1 : Learning Flow experience**	y1 : Balance of Tasks and Skills:	*7*	50	**0.81**	0.96	**0.87**	0.90
		y2 : Integration of Awareness and Action	*6*		**0.74**		**0.76**	
		y3:Purposefulness and Feedback	*6*		**0.80**		**0.85**	
		y4:Attentional Focus	*9*		**0.82**		**0.83**	
		y5:self-forgetfulness	*7*		**0.83**		**0.89**	
		y6:Sense of Control	*8*		**0.85**		**0.87**	
		y7 : Enjoyment of the Experience	*7*		**0.84**		**0.89**	

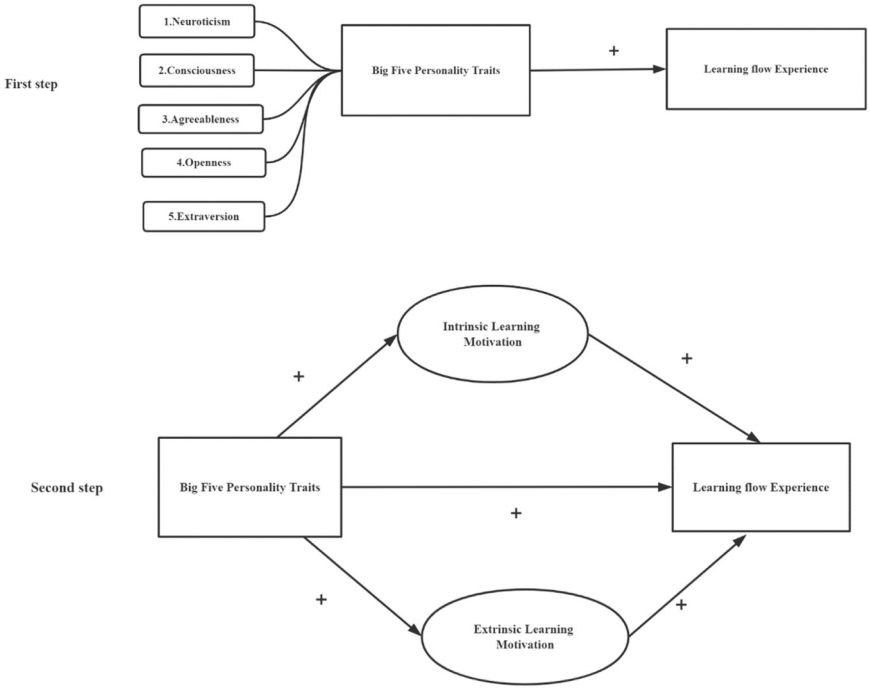


Fig. 1. The analysis process of this study.

dimensions of learning motivation.

To ensure the rigor of the analysis process, the results section is divided into two separate sections. The first section focuses on the development and sources of the scale, encompassing statistical analyses such as general descriptive statistics (mean and standard deviation), reliability analysis, and validity analysis. Following recent scholarly recommendations, a two-stage approach was adopted. Initially, measurement factors were examined, and necessary adjustments were made to enhance measurement accuracy and control errors. Subsequently, correlation analysis was conducted. The second section delves into the statistical analyses involved in scale development, including correlation analysis, robust regression analysis, and mediating role analysis. Causality was investigated using a two-stage goodness-of-fit test and modification. Finally, the significance of each estimated parameter and its potential causal structure were analyzed.

Overall, the research procedure entailed rigorous translation, careful sample selection, and thorough statistical analysis, ensuring

Table 3
Summary of Descriptive Statistics.

Variables			*Max-value*	*Min-value*	*Median value*	*St. Deviation*	*Number of variables in the questions*	*Number of samples*
(1–5):Personality traits	*X1. Neuroticism.*		48	18	35.5	5.8	8	450
	X2. Consciousness		48	18	35.0	5.9	8	
	X3. Agreeableness		45	8	25.6	6.8	8	
	X4. Openness		48	16	34.5	6.5	8	
	X5. Extraversion		46	15	31.7	6.3	8	
(6–7):Learning motivation	***M1. Intrinsic Learning Motivation***	m1 : Challenging:	48	35.6	6.1	6.1	8	
		m2 : Enthusiasm	36	27.1	4.5	4.5	6	
	M2. Extrinsic Learning Motivation	m3 : Choosing simple tasks	34	23.1	3.7	3.7	5	
		m4 : Relying on others to evaluate.	12	8.9	2.0	2.0	2	
		m5 : Seeking rewards	30	21.3	4.3	4.3	5	
(8–14):*Learning Flow experience*	y1 : Balance of Tasks and Skills:		42	15	31.1	5.3	7	
	y2 : Integration of Awareness and Action		36	9	24.2	5.2	6	
	y3:Purposefulness and Feedback		36	12	26.5	4.9	6	
	y4:Attentional Focus		54	16	37.8	7.5	9	
	y5:self-forgetfulness		42	7	26.7	6.8	7	
	y6:Sense of Control		48	10	31.9	7.5	8	
	y7 : Enjoyment of the Experience		42	13	30.8	6.0	7	

the reliability and validity of the study's findings.".

3.4. Data analysis tools

We collected 1350 valid questionnaires using previously described questionnaires and analyzed the data using SPSS version 26.0. Descriptive statistics and correlations were calculated to explore the relationships between variables. To test the significance of mediating effects, we utilized Baron and Kenny's (1986) causality stepwise regression test, a commonly used method for testing mediations (Wen & Ye, 2014). We conducted regression analysis and mediation analyses using the SPSS program (https://spssau.com/).

4. Results

4.1. Descriptive statistics

Descriptive statistics is an essential part of initial data analysis, providing a basis for comparing variables using inferential statistical tests (Kaur et al., 2018). In this study, we conducted descriptive statistical analyses to explore the association between personality traits, learning motivation, and learning flow experience among university students. The results are presented in Table 3.

Table 3 shows that college students have a balanced distribution of personality traits, scoring high on the dimensions of Agreeableness, Conscientiousness, and Openness, moderate on Extraversion, and low on Neuroticism. In other words, college students outperformed the theoretical average. The total motivation score and its two dimensions were above the intermediate level. The total learning flow experience scores and their scores on the seven dimensions were slightly above the theoretical median, with the overall level being moderately high (Ye, 2013).

4.2. Significance test of differences

The present study used the Independent Samples t-test provided by SPSS to evaluate the differences in demographic variables, including the Big Five personality traits, learning motivation, and levels of learning flow experiences. Factors such as gender, major, home location, being an only child or not were also tested separately. The results are presented in Table 4.

Table 4
Summary of Independent Samples t-test.

Gender difference : (Mean ± standard deviation)	X1 : Neuroticism	X2 : Consciousness	X3 : Agreeableness	X4 : Openness	X5 : Extraversion	M1 : Intrinsic Learning Motivation	M2 : Extrinsic Learning Motivation	Y : Learning Flow experience
Male (n = 193)	37.49 ± 5.37	36.89 ± 5.47	28.19 ± 6.48	36.35 ± 5.69	32.99 ± 5.71	66.96 ± 8.52	97.53 ± 12.48	224.86 ± 31.77
Female (n = 257)	32.96 ± 5.41	32.41 ± 5.44	22.10 ± 5.64	31.96 ± 6.59	30.07 ± 6.56	56.98 ± 9.36	84.14 ± 13.85	187.45 ± 34.41
t	*-8.815*	*-8.627*	*-10.416*	*-7.412*	*-4.928*	*-11.782*	*-10.748*	*-11.927*
p	*0.000* * *	*0.000* * *	*0.000* * *	*0.000* * *	*0.000* * *	*0.000* * *	*0.000* * *	*0.000* * *
Major difference	**X1**	**X2**	**X3**	**X4**	**X 5**	**M 1**	**M 2**	**Y**
Arts and History (n = 196)	32.69 ± 5.29	32.11 ± 5.55	23.61 ± 6.55	31.30 ± 6.29	28.95 ± 6.02	57.81 ± 9.44	83.86 ± 13.49	188.75 ± 35.54
Science and Engineering (n = 254)	37.75 ± 5.24	37.17 ± 5.15	27.10 ± 6.66	36.92 ± 5.46	33.89 ± 5.55	66.44 ± 9.05	97.91 ± 12.43	224.30 ± 31.68
t	-10.115	-9.996	-5.543	-10.143	-9.024	-9.85	-11.454	-11.19
p	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *
Family location difference	**X1**	**X2**	**X3**	**X4**	**X 5**	**M 1**	**M 2**	**Y**
Rural (n = 196)	33.47 ± 5.46	32.87 ± 5.19	23.20 ± 6.46	32.51 ± 6.30	30.22 ± 6.35	58.79 ± 9.40	85.93 ± 13.33	193.67 ± 34.57
Urban (n = 253)	38.23 ± 5.16	37.68 ± 5.63	28.67 ± 6.03	37.02 ± 5.76	33.70 ± 5.55	67.73 ± 8.82	99.38 ± 12.71	228.45 ± 32.33
t	-9.391	-9.398	-9.254	-7.903	-6.179	-10.276	-10.834	-10.884
p	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *
Only child or not difference	**X1**	**X2**	**X3**	**X4**	**X 5**	**M 1**	**M 2**	**Y**
No (n = 210)	33.59 ± 5.39	32.57 ± 5.19	23.48 ± 6.14	32.10 ± 6.04	29.76 ± 6.15	58.21 ± 8.88	85.34 ± 13.23	190.60 ± 32.57
Yes (n = 239)	37.78 ± 5.51	37.71 ± 5.42	27.98 ± 6.81	37.19 ± 5.84	34.00 ± 5.58	67.80 ± 9.08	99.16 ± 12.61	229.63 ± 32.13
t	-8.141	-10.262	-7.362	-9.058	-7.621	-11.305	-11.304	-12.762
p	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *	0.000 * *

Firstly, the t-test was conducted to evaluate university students' personality traits based on their gender differences. The results showed that the t-values for the Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion personality traits were significant for gender differences. Specifically, the Neuroticism personality trait ($t = -8.815$, $p = 0.000$) showed a 0.01 level of significance, with the mean for females (32.96) being substantially higher than that of males (37.49). On the other hand, male college students showed more obvious Consciousness($t = -8.627$, $p = 0.000$), Agreeableness($t = -10.416$, $p = 0.000$), Openness ($t = -7.412$, $p = 0.000$), and Extraversion ($t = -4.928$, $p = 0.000$) personality traits than female college students.

Secondly, Table 4 depicted the differences in personality traits based on different family locations. The results showed a 0.01 level of significance for the Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion personality trait ($t = -10.115$, -9.996 , -5.543 , -10.143 , -9.024; $p = 0.000$), with the mean for Arts and History (32.69 , 32.11 , 23.61 , 31.30 , 28.95) being significantly lower than that of Science and Engineering (37.75 , 37.17 , 27.10 , 36.92 , 33.89).

Thirdly, the study examined the differences in Big Five personality traits between different family locations. Five personality traits, Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion personality trait, showed significant differences ($p < 0.01$) between family locations, indicating differences in these traits between the samples. The results showed a 0.01 level of significance for the Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion personality trait (t-9.391 , -9.398 , -9.254 , -7.903 , -6.179; $p = 0.000$), with the mean for rural (33.47 , 32.87 , 23.20 , 32.51 , 30.22) being significantly lower than that of urban family students (38.23 , 37.68 , 28.67 , 37.02 , 33.70).

Fourthly, Table 4 showed that the Neuroticism and Conscientiousness personality traits were not significant ($p > 0.01$) for whether the student was an only child, indicating consistency in these traits regardless of whether or not the student was an only child. The results showed a 0.01 level of significance for the Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion personality trait ($t = -8.141$, -10.262 , -7.362 , -9.058 , -7.621; $p = 0.000$), with the mean for non-only children (33.59 , 32.57 , 23.48 , 32.10 , 29.76) being significantly lower than that of Science and Engineering (37.78 , 37.71 , 27.98 , 37.19 , 34.00).

Overall, the results of this study suggest that there are significant differences in personality traits, learning motivation, and learning flow experiences based on gender, major, family location, and being an only child or not. These findings can help educators and policymakers in designing more effective teaching strategies and support systems for students based on their individual differences.

4.3. Scatter plot

Scatter plots are commonly employed to analyze the correlation between quantitative data, specifically to examine the relationship between variables X and Y. These plots serve as valuable tools in exploratory studies, enabling the visualization of data relationships. In this study, Fig. 2 illustrates the scatterplot depicting the connection between internal and external learning motivation, and the learning immersion experience for participants based on the Big Five personality traits. Each row in the figure represents a distinct scatterplot, with the degree of learning immersion experience displayed on the y-axis at the end of the row. Likewise, each column represents a specific personality trait, with the corresponding scatterplots exhibiting the trait on the x-axis at the top of the column.

4.4. Spearman correlation analysis

Correlation analysis is a statistical tool that helps to investigate the relationship between quantitative variables, including the degree of closeness of the relationship. It is often used before regression analysis, as the logical relationship between correlation analysis and regression analysis is that a correlation needs to exist before a regression relationship can be established (ÇAPIK, 2020). The Pearson correlation coefficient is typically used for quantitative data that follows a normal distribution, while the Spearman

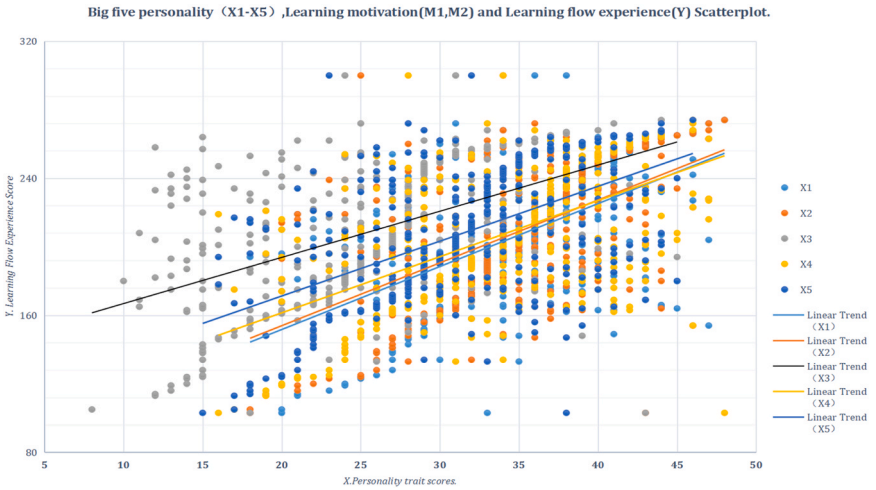


Fig. 2. Big five personality (X1-X5) ,Learning motivation(M1,M2) and Learning flow experience(Y) Scatterplot.

correlation coefficient is used for data that do not satisfy the normal distribution assumption (Minitab, 2019). In this study, the researcher checked the normal distribution of the three variables of interest by examining the eight groups of data (1−5), 6–7, and 8 shown in Table 4 of the.

Spearman Correlation Analysis.

The researcher found that the data were not normally distributed; therefore, the Spearman correlation coefficient was used to investigate the correlation between the variables. The following findings were obtained from the investigation.

In terms of the relevance of the variables, the results shown in Table 5 indicated that intrinsic learning motivation was significantly positively correlated with personality traits such as Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion ($r = 0.545$, $p < 0.01$; $r = 0.591$, $p < 0.01$; $r = 0.528$, $p < 0.05$; $r = 0.547$, $p < 0.01$; $r = 0.488$, $p < 0.01$). This suggests that there is a significant positive correlation between intrinsic learning motivation and personality traits such as consciousness, agreeableness, openness, and extraversion.

Furthermore, the results indicated that extrinsic learning motivation was significantly positively correlated with personality traits such as Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion ($r = 0.745$, $p < 0.01$; $r = 0.780$, $p < 0.01$; $r = 0.544$, $p < 0.05$; $r = 0.881$, $p < 0.01$; $r = 0.819$, $p < 0.01$). This implies that there is a significant positive correlation between extrinsic learning motivation and personality traits such as neuroticism and consciousness.

Lastly, the results showed that learning flow experience was significantly positively correlated with personality traits such as Neuroticism , Consciousness, Agreeableness, Openness, and Extraversion, intrinsic learning motivation, and extrinsic learning motivation($r = 0.575$, $p < 0.01$; $r = 0.589$, $p < 0.01$; $r = 0.536$, $p < 0.01$; $r = 0.562$, $p < 0.01$; $r = 0.525$, $p < 0.01$;$r = 0.706$, $p < 0.01$; $r = 0.681$, $p < 0.01$; $r = 0.700$, $p < 0.01$). These correlations were significant at the 0.01 level, indicating that there is a significant positive correlation between learning flow experience and personality traits such as neuroticism, consciousness, agreeableness, intrinsic learning motivation, and extrinsic learning motivation.

Overall, the Spearman correlation analysis revealed significant positive correlations between the variables of interest, suggesting that there is a relationship between these variables.

4.5. Regression analysis

In this study, we employed a two-step regression analysis to examine the personality traits that most effectively predict the degree to which the learning process experience is influenced. To ensure the reliability of our findings, we utilized a widely accepted approach among scholars in recent years. Initially, we employed ordinary least squares (OLS) regression to derive the initial results. Subsequently, we employed robust regression analysis to validate the robustness and consistency of the findings (Anger et al., 2017; Pallant, 2020; Zhang et al., 2023).

4.5.1. OLS regression analysis

The study examined the influence of various factors on the learning flow experience (Y). The control variables included gender (male or female), major (**Science and Engineering** or Arts and History), whether the child was an only child (yes or no), and family location (urban or rural). The dummy variables consisted of school level, grade level, and monthly family income. OLS regression analysis was conducted separately to explore the most influential personality traits on Y, and the results are presented in Table 6. Table 6 displays the results of the OLS regression analysis, where X1, X2, X3, X4, X5, M1, and M2 were used as independent variables. The major variable was represented by dummy variables for Arts and History (1) and **Science and Engineering** (2). The analysis employed the Robust robust standard error regression method. The specific analysis revealed that the regression coefficient for Science and Engineering majors (2.0) was 9.384, indicating a significant positive effect on Y at the 0.01 level of significance ($t = 3.322$, $p = 0.001 < 0.01$). This means that students pursuing **Science and Engineering** majors had a more favorable learning flow experience compared to major in Arts and History students.

Additionally, the regression coefficient values for personality traits X1, X2, X4, and X5 were 0.760, 0.030, − 0.515, and 0.509, respectively. However, none of these coefficients reached statistical significance ($t = 1.166$, $p = 0.244 > 0.05$; $t = 0.060$, $p = 0.952 > 0.05$; $t = -1.112$, $p = 0.266 > 0.05$; $t = 1.314$, $p = 0.189 > 0.05$). Thus, personality traits X1, X2, X4, and X5 were not found to have a significant effect on Y. On the other hand, the regression coefficient for X3 was 0.678 and demonstrated a 0.01 level of

Table 5
Spearman Correlation Analysis.

	X1	X2	X3	X4	X5	M1	M2	M	Y
X1	1								
X2	0.820 * *	1							
X3	0.252 * *	0.316 * *	1						
X4	0.812 * *	0.807 * *	0.216 * *	1					
X5	0.727 * *	0.711 * *	0.100 *	0.829 * *	1				
M1	0.545 * *	0.591 * *	0.528 * *	0.547 * *	0.488 * *	1			
M2	0.754 * *	0.780 * *	0.544 * *	0.881 * *	0.819 * *	0.670 * *	1		
M	0.549 * *	0.600 * *	0.567 * *	0.545 * *	0.494 * *	0.947 * *	0.685 * *	1	
Y	**0.575****	**0.589****	**0.536****	**0.562****	**0.525****	**0.706****	**0.681****	**0.700****	1

*$p < 0.05$, * * $p < 0.01$.

Table 6
OLS regression analysis results (n = 450).

	Regression coefficient	Standard Error	t	p	95% CI
Constant	20.093	7.760	2.589	***0.010*** * *	4.884 ~ 35.301
Arts and History major【Reference group】	-	-	-	-	-
Science and Engineering major	9.384	2.825	3.322	***0.001*** * *	3.847 ~ 14.921
X1	0.760	0.652	1.166	0.244	-0.517 ~ 2.037
X2	0.030	0.494	0.060	0.952	-0.938 ~ 0.998
X3	0.678	0.198	3.420	***0.001*** * *	0.289 ~ 1.067
X4	-0.515	0.464	-1.112	0.266	-1.424 ~ 0.393
X5	0.509	0.388	1.314	0.189	-0.250 ~ 1.269
M1	1.244	0.226	5.500	***0.000*** * *	0.800 ~ 1.687
M2	0.672	0.171	3.923	***0.000*** * *	0.336 ~ 1.008
R^2	0.594				
Adjustment R^2	0.588				
F	F (7442)= 161.419,p = 0.000				
D-W Value	1.075				

Dependent variable: Y. Learning flow experience summation.
* $p < 0.05$ * * $p < 0.01$.

significance (t = 3.420, p = 0.001 <0.01), suggesting a significant positive relationship between X3 and Y.(Learning flow experience summation).

Furthermore, the regression coefficient for mediating variable M1 (Learning Intrinsic Motivation) was 1.244, showing a 0.01 level of significance (t = 5.500, p = 0.000 <0.01). This implies that M1 had a significant positive influence on Y. Similarly, the regression coefficient for M2 (Learning Extrinsic Motivation) was 1.244, indicating a significant positive influence on 'Y' at the 0.01 level of significance (t = 5.500, p = 0.000 <0.01).

In summary, the analysis revealed three findings: .

a) Students from different majors had varying levels of learning flow experiences.
b) The personality trait X3 had the strongest influence on Y and produced a significant positive effect.
c) Internal motivation (M1) and external motivation (M2) both significantly influenced Y positively.

4.5.2. Robust regression analysis

The Table 7 presents the results of a robust regression analysis (M estimation method) conducted to examine the relationship between independent variables (X1, X2, X3, X4, X5, M1, M2) and the dependent variable (total Y), while controlling for different majors. The regression coefficient for major difference was found to be 12.932 (t = 6.838, p = 0.000 <0.01), indicating a significant positive association between different major and total Y. On the other hand, the regression coefficient for X1 was − 0.044 (t = −0.162, p = 0.872 >0.05), suggesting that X1 does not have a significant impact on total Y. Similarly, the regression coefficient for X2 was 0.265 (t = 0.973, p = 0.330 >0.05), indicating that X2 does not have a significant effect on total Y. The regression coefficient for X4 was − 0.139 (t = 0.566, p = 0.572 >0.05), implying that X4 does not have a significant influence on total Y. Additionally, the regression coefficient for X5 was 0.359 (t = 1.677, p = 0.094 >0.05), suggesting that X5 does not affect total Y.

Conversely, the regression coefficient for X3 was 0.605 (t = 5.259, p = 0.000 <0.01), indicating a significant positive relationship between X3 and total Y. The regression coefficient for M1 was 1.344 (t = 11.806, p = 0.000 <0.01), implying a significant positive effect of M1 on total Y. Similarly, the regression coefficient for M2 was 0.825 (t = 10.261, p = 0.000 <0.01), indicating a significant

Table 7
Robust regression analysis results (n = 450).

	Regression coefficient	Standard Error	t	p	95 % CI	R^2	Adjustment R^2	F
Constant	-1.671	5.852	-0.286	0.775	-13.141 ~ 9.799	0.584	0.576	F (8441)= 77.352, p = 0.000
Different Major	12.932	1.891	6.838	**0.000** **	9.225 ~ 16.638			
X1	-0.044	0.274	-0.162	0.872	-0.582 ~ 0.494			
X2	0.265	0.273	0.973	0.33	-0.269 ~ 0.800			
X3	0.605	0.115	5.259	**0.000** **	0.380 ~ 0.830			
X4	-0.139	0.246	-0.566	0.572	-0.622 ~ 0.343			
X5	0.359	0.214	1.677	0.094	-0.061 ~ 0.778			
M1	1.344	0.114	11.806	0.000**	1.121 ~ 1.568			
M2	0.825	0.08	10.261	0.000**	0.667 ~ 0.982			

* $p < 0.05$.
** $p < 0.01$.

Table 8
Table on the mediating effect of intrinsic learning motivation on Agreeableness and learning flow experience.

	Y					M1					M2					Y				
	B	Standard error	*t*	*p-value*	*β*	*B*	Standard error	*t*	*p-value*	*β*	*B*	Standard error	*t*	*p-value*	*β*	*B*	Standard error	*t*	*p-value*	*β*
Constants	139.869 * *	6.036	23.174	0.000	-	43.340 * *	1.603	27.041	0.000	-	60.307 * *	2.198	27.432	0.000	-	15.307	8.006	1.912	0.057	-
X3	2.695 * *	0.228	11.822	0.000	0.488	0.756 * *	0.061	12.490	0.000	0.508	1.231 * *	0.083	14.821	0.000	0.574	0.332	0.212	1.564	0.119	0.060
M1																1.395 * *	0.159	8.794	0.000	0.375
M2																1.063 * *	0.116	9.198	0.000	0.413
R^2	0.238					0.258					0.329					0.576				
Adjustment R^2	0.236					0.257					0.327					0.573				
F-value	*F* (1448)= 139.771,*p* = 0.000					*F* (1448)= 156.004,*p* = 0.000					*F* (1448)= 219.654,*p* = 0.000					*F* (3446)= 201.581,*p* = 0.000				

Note：B=Non standardized regression coefficient；t = B/Standard error；β = standardized regression coefficient.
* $p < 0.05$ * * $p < 0.01$.

positive influence of M2 on total Y. Notably, these findings are consistent with the scatter plot in Fig. 2 and align with the results of the OLS regression. Consequently, the conclusion drawn from the analysis is robust and reliable.

4.6. Mediation analysis

The present study aimed to identify the personality traits that have the most significant influence on the learning flow experience, and we hypothesized that different personality traits affect the learning flow experience through the mediating variable of "students' learning motivation," as measured by the Learning Motivation Questionnaire. Using stepwise regression analysis, we found that university students with higher Agreeableness scores had a higher level of learning flow experience. However, we also proposed that personality traits affect the level of learning flow experience through their correlation with students' learning motivation. To investigate how personality traits and learning motivation directly or indirectly influence the learning flow experience using different coping styles (intrinsic or extrinsic motivation), we conducted a mediation analysis.

Based on our theoretical framework, we tested whether Intrinsic Learning Motivation mediates the effect of Agreeableness personality traits on the learning flow experience. As Agreeableness (X) was not correlated with extrinsic motivation, only intrinsic motivation (M) was chosen as the mediating variable. The mediating effect study involves several indicators of relevance, which are described as follows. First, c denotes the regression coefficient when X is on Y (when there is no mediating variable M in the model), and c denotes the total effect. Second, a denotes the regression coefficient when X is on M, b denotes the regression coefficient when M is on Y, and ab is the product of a and b, denoting the mediating effect. Third, c' denotes the regression coefficient when X is on Y (when there is a mediating variable M in the model), and c' denotes the direct effect. Fourth, the effect is fully mediated if a and b are significant, and c' is insignificant. Fifth, the effect is partially mediated if a and b are significant, c' is significant, and ab has the same sign as c'.

Tables 8–9 summarizes the mediating effect of Intrinsic Learning Motivation on Agreeableness and the learning flow experience. The results of the mediating effect are shown in Fig. 3, which includes four models involved in the analysis of the mediating effect as follows:

$Y = 139.869 + 2.695 *X3$

$M1 = 43.340 + 0.756 *X3$

$M2 = 60.307 + 1.231 *X3$

$Y = 15.307 + 0.332 *X3 + 1.395 *M1 + 1.063 *M2$

We refined Fig. 1 to obtain the following Fig. 3**.**

In summary, our findings demonstrate that personality trait Agreeableness (X) can influence the learning flow experience (Y) degree through Intrinsic Learning Motivation (M).

5. Discussions

Learning flow experience is a complex behavior affected by different personality and environmental factors (Custodero, 2005; Oliveira et al., 2021). Considering previous theories about the big five personality traits (Komarraju & Karau, 2005; Komarraju et al., 2009) and learning motivation e.g. (Apostolov & Geldenhuys, 2022; Hazrati-Viari et al., 2012) , the present study explores the relationship between big five personality and college students' learning flow experience. In addition, it examines the mediation mechanisms underlying this relationship by examining the potential mediating role of intrinsic learning motivation and extrinsic learning motivation.

This study is based on the Big five personality factor theoretical model. It uses the Chinese Big Five personality questionnaire short version of (Wang et al., 2010). To examine contemporary university students' personality traits to understand the overall distribution of personality traits among university students. The study results showed that university students' overall distribution of personality traits was balanced. They scored high on the dimensions of Agreeableness, Conscientiousness and Openness, moderate on Extraversion , and relatively low on Neuroticism. This variation in scores is in line with the variation in the normative scores of university students are given by Wang et al. (2010)in the "Chinese Big Five Personality Questionnaire Short Version". It is also consistent with the variation in scores in study of (Mammadov, 2022; Smith-Woolley et al., 2019). The highest scores were on the trait of Agreeableness, reflecting the general characteristics of young university students who are more capable of "loving" and place more importance on cooperation and interpersonal and harmonious relationships. The higher scores were for Conscientiousness and Openness, with

Table 9
Summary of intermediary role test results.

	c Total effect	a	b	a*b Intermediary effect	a*b (Boot SE)	a*b (*z-value*)	a*b (*p-value*)	a*b (95 % BootCI)	c' Direct effects	Test Conclusion
X3 = >M1 = >Y	2.695**	0.756**	1.395**	1.054	0.035	29.934	0.000	0.124 ~ 0.263	0.332	Fully mediated
X3 = >M2 = >Y	2.695**	1.231**	1.063**	1.309	0.037	35.705	0.000	0.166 ~ 0.311	0.332	Fully mediated

* $p < 0.05$.
** $p < 0.01$.

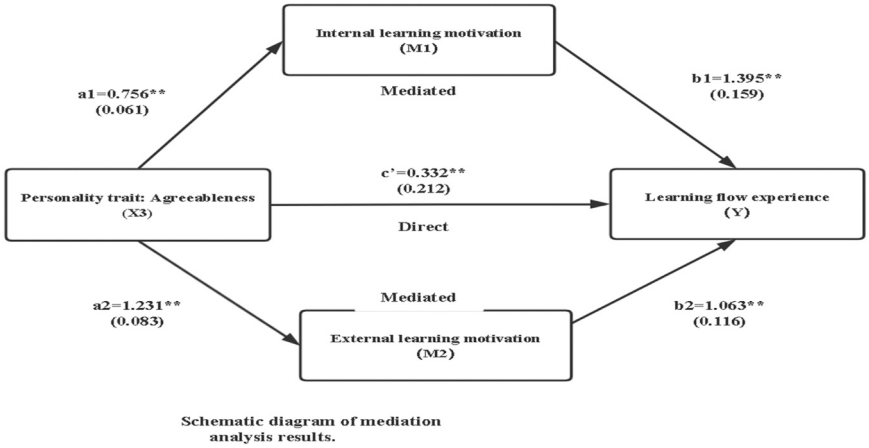


Fig. 3. Schematic diagram of mediation analysis results.

Conscientiousness representing a more cautious approach to controlling, managing, and regulating our impulses, and Openness reflecting the more open and creative cognitive style of today's university students. The average scores for Extraversion reflect the even distribution of introverted and extroverted personalities in the population. Neuroticism scores are the lowest of the five dimensions, reflecting contemporary university students' overall positive affect regulation and more stable emotions. From the results of the study, the total score of college students' learning motivation and its six dimensions are all higher than the theoretical median, and the overall level is moderately high for men, which indicates that the overall level of college students' learning motivation in China is high. This study's conclusion are the same as those of (Gao, 2020; Zhang, Shi, & Zhou, 2021) , but contrasts (Li, 2012; Zhao & Mei, 2016) finding that current Chinese university students are generally not sufficiently learning motivation. The study's findings indicate that the total learning flow experience scores of university students is slightly higher than the theoretical median, and the overall level is moderately male to high, which indicates that the overall level of learning flow experience of university students in China is high.

By the previous data analyses, we can confirm that H1-H3 are all valid; that means Big five personality traits are positively correlated with learning motivation, and Big five personality traits are positively correlated with learning flow experience. College students' motivation to learn positively correlates with the learning flow experience.

The results of the correlation analysis between personality traits and learning motivation of university students showed that the personality trait dimensions of university students were significantly and positively correlated with learning motivation. This is in line with (Clark & Schroth, 2010; Hazrati-Viari et al., 2012; Komarraju & Karau, 2005; Komarraju et al., 2009; Liu & Yang, 2014; L. Zhang et al., 2019) regarding the correlation between personality traits and motivation to learn findings are consistent. In addition, this study explored the relationship between personality traits and intrinsic and extrinsic learning motivation, specifically: Consciousness, Agreeableness, Openness, and Extraversion were significantly and positively correlated with intrinsic learning motivation, this is consistent with the findings of (Komarraju et al., 2009); Neuroticism and Consciousness were significantly and positively correlated with extrinsic motivation. This is in line with (Jensen, 2015; Liang & Kelsen, 2018) Study results in agreement.

The results of the correlation analysis between personality traits and learning flow experience showed that university students' three personality trait dimensions were significantly and positively correlated with the learning flow experience. This is consistent with the findings of (Annalakshmi et al., 2020; Liang, 2013; Ye, 2013; Yu et al., 2020) and others regarding the correlation between personality traits and learning immersion experiences, Though research by Hager (2015) suggests that although individuals with higher personality traits may be more inclined to experience flow, neuroticism tendencies may hinder an individual's tendency to experience flow. In addition, this study explored the relationship between personality traits and learning flow experience; Neuroticism, Consciousness, and Agreeableness were significantly and positively correlated with learning immersion experience. Specifically, the personality traits Neuroticism, Consciousness, and Agreeableness significantly positively affected the immersion experience, meaning that university students with these three traits were likelier to have a flow experience.

The results of the correlation analysis between university students' learning motivation and learning flow experience showed that university students' intrinsic and extrinsic learning motivation was significantly and positively correlated with learning flow experience, which is consistent with the findings of (Hofferber, Basten, Großmann, & Wilde, 2016; Hong, Hwang, Tai, & Lin, 2017; Kong & Wang, 2021) and others on the correlation between motivation and learning flow experiences among university students are consistent. Specifically: 1. Intrinsic Learning Motivation was significantly and positively correlated with the learning flow experience, which is consistent with the findings of (Chang, Warden, Liang, & Lin, 2018; Hong, Hwang, Tai, & Lin, 2017). 2. Extrinsic Learning Motivation was significantly and positively correlated with the learning flow experience, This is consistent with the (Kong & Wang, 2021) study phenomenon.

Which personality trait most influences the learning flow experience? How does this personality trait influence the learning flow experience directly or indirectly through using different coping styles (intrinsic or extrinsic motivation)?

From model results from stepwise regression analysis, we found that university students with more Agreeableness person-identity personality traits had the highest learning flow experience. Also tested by mediating effects, the personality trait 'Agreeableness' (X3)

could influence learning flow experience through intrinsic learning motivation (M1) with an indirect effect a1 *b1 = 1.054,and through extrinsic learning motivation (M2) with an indirect effect a2 *b2 = 1.309, but the direct effect of personality trait 'agreeableness' (X3) directly influencing learning flow experience (Y) was only c'= 0.332.

6. Conclusion

Taking university students in China as participants, this study examined the mediating roles of learning motivation between the big five personality traits and learning flow experience. The research results support four proposed hypotheses:

College students' personality traits, Internal and extrinsic learning motivation, and learning flow experience differed significantly in demographic variables. Conscientiousness ， Agreeableness and Openness personality traits were correlated with intrinsic learning Motivation; Conscientiousness ， Agreeableness ， and Extraversion personality traits were correlated with extrinsic learning Motivation. （2）Internal and extrinsic learning Motivation were correlated with learning flow experience; Intrinsic learning Motivation was positively correlated with learning flow experience. (3) Conscientiousness and Agreeableness personality traits were correlated with extrinsic learning Motivation. (4) Learning motivation mediates the relationship between personality traits and learning flow experience. Agreeableness predicts the learning flow experience significantly; Agreeableness has a significant positive predictive effect on the learning flow experience. In the summation of the learning flow experience, Intrinsic and Extrinsic learning Motivation would have a considerable positive predictive effect. Furthermore, the association between Agreeableness and learning flow experiences were mediated by Intrinsic Learning Motivation.

The current study makes an innovative contribution to understanding the relationship between the big five personality traits and learning flow experience, as it emphasizes the co-mediating role of learning motivation in the relationship between the big five personality traits and learning flow experience. In a flow environment involving learning, teachers or parents may choose to enhance the learning flow experience for students with different personality traits by promoting intrinsic or extrinsic motivation for learning. In this study, the relationship between different personality traits and learning motivation in a flow learning experience environment is theoretically explored in depth, and more scientific findings were developed. It provides a valuable reference for further exploring the relationship between different personality traits and learning motivation in an authentic learning flow experience environment.

The current research has several limitations. The current study has several limitations. Firstly, it uses a research design with all Chinese participants. The survey data used in this study led us to draw no broader conclusions about globalization. The conclusions may only apply to China or countries with equivalent education systems, which suggests the need for more schools in other countries in future studies. Secondly, in terms of research tools and methods, this study only used questionnaires to conduct the study and did not utilize other methods, such as interviews and experiments, to help research. In future studies, scholars can use various methods for further research. Again, in terms of research variables. In the literature on learning flow experiences, a single influential variable, motivation, was introduced to explore its possible role as a partial mediator between personality traits and immersion learning experiences. However, there are many other antecedent variables of learning flow experiences. Then, in terms of outcome analysis. Most of this study only explored the five dimensions of personality traits of university students and the overall and individual differences in the three variables of intrinsic motivation to learn, extrinsic motivation and learning immersion experience. However, it did not dig deeper and explore in detail all the dimensions of the variables involved, which needs further exploration in future studies. Finally, in terms of the application of the findings. It is worth thinking about how to enhance learning immersion experiences in practice, how to shape and develop students' ideal personalities, how to increase students' extrinsic learning motivations and, in some aspects, intrinsic learning motivation ， and how to take steps to increase the depth and frequency of students' learning flow experiences concerning their personality traits and different learning motivation.

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CRediT authorship contribution statement

Yihui Tian: Conceptualization. **Yihui Tian:** Methodology and Formal analysis. **Yihui Tian** and **LingJun Ou:** Software and Data curation. **LingJun Ou:** Investigation, Resources, Writing – review & editing, Visualization, Supervision. All authors have Writing – original draft preparation, Validation, read, and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare no conflict of interests in this research.

Data availability

Data will be made available on request.

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Appendix-1. The detailed descriptions of the corresponding factors of three questionnaires

Questionnaire 1: Short Version of the Chinese Big Five Personality Questionnaire.

These factors collectively represent a comprehensive framework for understanding and assessing personality traits, and they have been extensively studied and validated within the field of personality research (Wang et al., 2010; X Zhang et al., 2019).

Questionnaire 1: Short Version of the Chinese Big Five Personality Questionnaire.

These factors collectively represent a comprehensive framework for understanding and assessing personality traits, and they have been extensively studied and validated within the field of personality research (Wang et al., 2010; X. Zhang, Wang, et al., 2019).

Variables	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
A1- **Neuroticism** (N) *This factor encompasses negativity and perception*	3. I often feel scared. 6. Sometimes I feel like I am useless. 11. I often relate to myself a careless remark made by someone else. 16. When faced with pressure, I feel like I am going to collapse. 21. I often worry about things that don't matter. 26. I often feel unsettled inside. 31. I often fear that something bad is going to happen. ***36.*** *I rarely feel down or depressed.*	**0.78**	**0.77**	8	450
A2–**Consciousness** (C) *Conscientiousness entails traits like fairness and restraint in one's actions.*	3. Once I have set a goal, I will work hard to achieve it. 7. I often make decisions only after careful consideration. 12. People think I am a prudent person. 17. I like to have things planned out at the beginning. 22. I work or study extremely hard. 27. I am a man who gives his best to his work. ***32****. When it comes to work, I often just try to get by.* 37. Being logical and organized is a characteristic of mine.	**0.76**	**0.84**	8	
A3–Agreeableness (A) *Agreeableness comprises qualities such as being cheerful, altruistic, and having a contagious positive demeanour.*	3. Most people are well-intentioned. ***8.*** *I do not really care if people are treated unfairly.* 13. Other people's pain is none of my business. 18. I am the kind of person who just takes care of myself and doesn't worry about others. 23. Although there are some scammers in society, most people are still trustworthy. 28. I often feel sad when people tell me about their misfortunes. 33. Despite the darkness of human society (e.g., war, sin, fraud), I believe human nature is generally good. 38. I often feel sorry for those who have suffered misfortune.	**0.81**	**0.89**	8	
A4–Openness (O) *Openness includes characteristics such as straightforwardness, creativity, and receptiveness to new ideas.*	4. My mind is often filled with vivid images. 9. I am a risk-taker and out-of-the-box person. 14. I like adventure. 19. I have a strong curiosity about many things. 24. I have a spirit of adventure that no one else has. 29. I am eager to learn new things.	**0.83**	**0.85**	8	

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Variables	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
	34. I have a vivid imagination. 39. I am open and receptive to new things, perspectives, and ideas.				
A5– **Extraversion** (E) *Extraversion encompasses attributes such as enthusiasm and dynamism in social interactions.*	***5.*** *I get bored with parties with lots of people.* 10. I often take the initiative and have fun at lively parties. ***15.*** *I try to avoid crowded parties and noisy environments* 20. There's usually no chill when I'm around. 25. In a group, I want to be in a leadership position. 30. People perceive me as warm and friendly. 35. I enjoy attending social and recreational gatherings. 40. I prefer to be a leader rather than a follower. 5. I get bored with parties with lots of people.	**0.72**	**0.76**	8	

Note: Questions in ***'bold italics'*** need to be scored in reverse.

Questionnaire 2: Learning Motivation Questionnaire.

The Learning Motivation Scale can be clearly distinguished into two subscales for measuring endogenous and exogenous motivation, with the former comprising two dimensions of challenge and enthusiasm and the latter comprising four dimensions of dependence on others for evaluation, choice of simple tasks, concern for interpersonal competition and pursuit of rewards(Chi & Xin, 2006).

Variables	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
B1– Challenging: *the tendency to do one's best with challenging work and important projects.*	2. The more difficult the problem, the more willing I am to try to solve it. 4. I hope that the work I do will provide me with the opportunity to increase my knowledge and skills. 6. I like to think independently to solve problems. 9. I am clear about the goals I want to achieve in terms of academic performance. 12. I enjoy delving into problems that are completely new to me. 14. I will be more satisfied when I can set my own goals. 22. I enjoy trying to solve complex problems. 23. For me, it was important to have the opportunity to express myself.	**0.81**	**0.83**	8	450
B2–Enthusiasm: *the individual's ability to devote themselves to their work and to enjoy it for what it is*	7. Whatever the outcome of what I do, if I can feel satisfied that I am getting a new experience. 10. I do a lot of things driven by curiosity. 16. It is important to me to be able to do work that I enjoy. 18. I don't care so much about marks and rewards if I'm doing something I'm happy to do. 19. I enjoy doing interesting work that takes my mind off things 26. For me, the most important thing is to love what I do.	**0.73**	**0.79**	6	
B3–Choosing simple tasks	1. I prefer to have someone at work who sets clear goals for me. 3. I am clear that my goal or aim is to achieve good grades. 5. For me, success means doing better than others. 8. I like tasks or assignments that are simple and straightforward	**0.7**	**0.7**	5	

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Variables	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
	15. For me, what I can win is the main thing that drives me to work hard.				
B4–Relying on others to evaluate.	24. I want to know how well I can really do academically. 25. I want people to find out how good I'm going to be academically.	**0.73**	**0.5**	2	
B5– Seeking rewards	11. I am more concerned not with what I do but what I get out of it. 13. I care a lot about how people react to my views. 17. I like to do work or tasks that have noticeably clear procedural steps. 20. Being able to win the recognition and appreciation of others is the main motivation for me to work hard. 21. Whatever I do, I always expect to be paid or rewarded for it.	**0.72**	**0.74**	5	

Questionnaire 3: Learning Flow Experience Questionnaire.

Based on the relevant theoretical support provided by (Ye, 2013)and(Ye, 2018),this paper proposes a comprehensive framework for understanding university students' learning immersion experience. This framework comprises seven distinct dimensions that contribute to the overall immersion experience. These dimensions include the balance of tasks and skills, purpose and feedback, integration of awareness and action, concentration, self-forgetfulness, sense of control, and enjoyment of the experience. By considering these dimensions, we aim to enhance our understanding of the factors that influence students' engagement and immersion in the learning process.

The seven dimensions/ Variables of questionnaire 3.	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
y1–Balance of Tasks and Skills: *The individual's experience regarding the balance between challenges and skills encompasses various perceptions and the confidence derived from their ability to successfully tackle a range of challenges.*	1. I feel competent in meeting the requirements of classroom learning. 8. I feel competent enough to handle difficult questions from my teacher. 20. I always have clear goals and objectives for my studies. 31. I clearly understand the teaching and learning objectives. 37. I am willing to take on challenges and try new things in my studies. 41. I feel confident in my ability to handle learning challenges. 48. The tasks and skills I challenge myself with are at the appropriate level.	**0.809**	**0.874**	7	450
y2–Integration of Awareness and Action : *The student possesses the capability to fully immerse themselves in an activity without evaluating or questioning their participation in it. Their awareness and actions are harmoniously aligned.*	2. I can think and work quickly and efficiently. 16. I am open-minded and able to learn more efficiently. 18. I have a laser-like focus on what I am paying attention to. 23. My body movements are minimal and purposeful while studying. 32. I have a serious and focused expression while studying. 33. I am able to maintain a serious or neutral facial expression while studying.	**0.739**	**0.755**	6	450
y3–Purposefulness and Feedback : *Tasks that possess a clear purpose are the ones that engender an immersive experience for the individual. Furthermore, the individual's immersion is facilitated by timely feedback as they progress through each activity.*	3. I have clear goals and know what I want to achieve. 9. I recognize and appreciate my growing knowledge and skills.	**0.804**	**0.852**	6	450

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The seven dimensions/ Variables of questionnaire 3.	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
	17. I stay focused on the learning material without getting distracted. 42. I feel like my brain is expanding to accommodate more knowledge. 43. I am in a good mood and quickly become engaged in my studies. 44. I feel confident in my abilities and enjoy learning new things.				
y4–Attentional Focus: : *During participation in the activity, the individual becomes so engrossed that their sense of time becomes distorted or altogether lost. They can fully engage with the environment at the activity site.*	4. I can concentrate fully on my studies. 7. I feel fully immersed in my studies and lose track of time. 11. My mind is highly focused on the content of my studies. 12. I am fully absorbed in the study question and material. 14. I am unaware of anything else going on around me. 28. I sometimes feel like there is not enough time to study. 35. I am aware of the passage of time while studying. 40. I become so focused on my studies that I forget about time and other things. 46. I am fully focused on the current study material without any distractions.	**0.822**	**0.83**	9	450
y5–Self-forgetfulness : *The participant exhibits complete concentration on the activity, disregarding their surroundings, including other individuals or objects. They exist in a state where the whole world appears to revolve solely around them, unaffected by external factors.*	5. My body stays in one position without discomfort 13. I am able to tune out any distracting noise or interruptions. 15. My attention is solely focused on the learning material. 25. I feel like I am in my own world while studying. 34. I sometimes become so absorbed in my studies that I talk to myself. 45. I do not feel the presence of others while studying. 50. I become fully immersed in my studies connected only to highly focused objects.	**0.828**	**0.885**	7	450
y6–Sense of Control: *The interaction between the individual and the activity they partake in becomes pivotal in their need to feel a sense of control over this interaction.*	6. I am not influenced by the behavior or opinions of others around me. 22. I am able to block out distractions and stay fully focused. 24. I am less disturbed by external distractions while studying. 29. I am so focused on my studies that I can't hear people talking to me 30. I pay close attention to what I am learning and what comes next. 39. I can ignore my surroundings even if there is a lot of noise. 47. I can study without being disturbed regardless of the environment. 49. I automatically filter out any irrelevant information while learning.	**0.848**	**0.871**	8	450
y7—Enjoyment of the Experience: *The student experiences a psychological state of pleasure and enjoyment in the process of learning. This pleasurable psychological experience stems directly from the current*	10. I feel a sense of happiness and fulfillment from my studies. 19. I am able to study for long periods without feeling tired.	**0.844**	**0.887**	7	450

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The seven dimensions/ Variables of questionnaire 3.	The composition items of each variable	Cronbach's alpha	KOM values	Number of items	Number of samples
activity in which the individual is engaged and is not influenced by external factors.	21. I feel happy and confident in my mind and body while studying 26. I feel calm and cheerful while studying. 27. I enjoy studying and find it fulfilling. 36. My body feels relaxed and at ease while studying. 43. I am in a good mood and quickly become engaged in my studies.				

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